

Prove the Jordan curve theorem: If C is a Jordan curve, i.e. the image of $\gamma : \mathbb{S}^1 \rightarrow \mathbb{R}^2$ which is injective and continuous, then $\mathbb{R}^2 \setminus C$ consists of two connected components and Γ is their common boundary.

We use several methods to prove this theorem.

Proof by Algebraic Topology

Recall some facts from algebraic topology:

- $H_i(\mathbb{S}^n) = \mathbb{Z}$ if $i = 0, n$, and $H_i(\mathbb{S}^n) = 0$ otherwise;
- By Poincaré duality, $H^i(\mathbb{S}^n) = \mathbb{Z}$ if $i = 0, n$ and $H^i(\mathbb{S}^n) = 0$ otherwise;
- $\tilde{H}_0(\mathbb{S}^2 - C) = \tilde{H}^0(C)$ by Alexander duality, where $C \cong \mathbb{S}^1$, hence $\tilde{H}_0(\mathbb{S}^2 - C) = \mathbb{Z}$ so $H_0(\mathbb{S}^2 - C) = \mathbb{Z} \oplus \mathbb{Z}$. Likewise $H_0(\mathbb{S}^2 - (C - \{x\})) = \mathbb{Z}$ since $C - \{x\} \cong \mathbb{R}$.

Now, view $\mathbb{S}^2$ as the one-point compactification of $\mathbb{C}$, then $H_0(\mathbb{S}^2 - C) = \mathbb{Z} \oplus \mathbb{Z}$, so $\mathbb{S}^2 - C$ has two connected components, denoted by D_+, D_- .

Since D_+, D_- are disjoint and open, clearly $\partial D_+, \partial D_- \subset C$. Recall the following lemma in general topology: if X, Y disjoint open, P connected and $P \cap X, P \cap Y \neq \emptyset$, then $P \cap \partial X \neq \emptyset$. Analogously, since $H_0(\mathbb{S}^2 - (C - \{x\})) = \mathbb{Z}$, $\mathbb{S}^2 - (C - \{x\})$ is connected, so let $X = D_+, Y = D_-$ and $P = \mathbb{S}^2 - (C - \{x\})$ we obtain $x \in \partial D_+, \partial D_-$ for all $x \in C$. Therefore $\mathbb{S}^2 - C$ has two connected components, with C as their common boundary.

By definition, $\mathbb{S}^2 = \mathbb{C} \cup \{\infty\}$, and we can assume $\infty \in D_+$. Let $U = D_+ - \{\infty\}$ and $V = D_-$, then U, V are open, disjoint, and still connected since ∞ is contained in a disk $O \subset D_+$. Hence U, V are the connected components of $\mathbb{R}^2 - C$, and clearly their boundaries are both C . $\square$

Proof Using Graph Theory

There is an interesting proof using the fact that $K_{3,3}$ is not a planar graph.

Lemma1: If C is a PL (Piecewise Linear) closed curve in $\mathbb{R}^2$, then $\mathbb{R}^2 \setminus C$ has exactly two connected components.

Proof: By taking $a \in \mathbb{R}^2 \setminus C$, and for each $x \in \mathbb{R}^2 \setminus C$, consider the parity of number of intersections of C with $[a, x]$ (segment from a to x), if $[a, x]$ passes a vertex or covers an edge, we view it as two intersections. We know that $\mathbb{R}^2 \setminus C$ is disconnected.

If q_1, q_2, q_3 lie in distinct components, take a small disk D such that $D \cap C$ is a line segment. Then take a tubular neighborhood of C , and walking along this neighborhood we obtain polygonal arcs in $\mathbb{R}^2 \setminus C$ sending q_i into D , a contradiction. $\square$

The unbounded component of $\mathbb{R}^2 \setminus C$ is $\text{ext}(C)$ and the bounded component is $\text{int}(C)$. Let $\overline{\text{ext}}(C) = C \cup \text{ext}(C)$ and $\overline{\text{int}}(C) = C \cup \text{int}(C)$.

Lemma2: Following Lemma 1, if P is a PL arc in $\overline{\text{int}}(C)$ joining $p, q \in C$ and otherwise disjoint with C . Let P_1, P_2 be the two arcs of C from p to q . Then $\mathbb{R}^2 \setminus (C \cup P)$ has 3 components, with boundaries $C, P \cup P_1, P \cup P_2$.

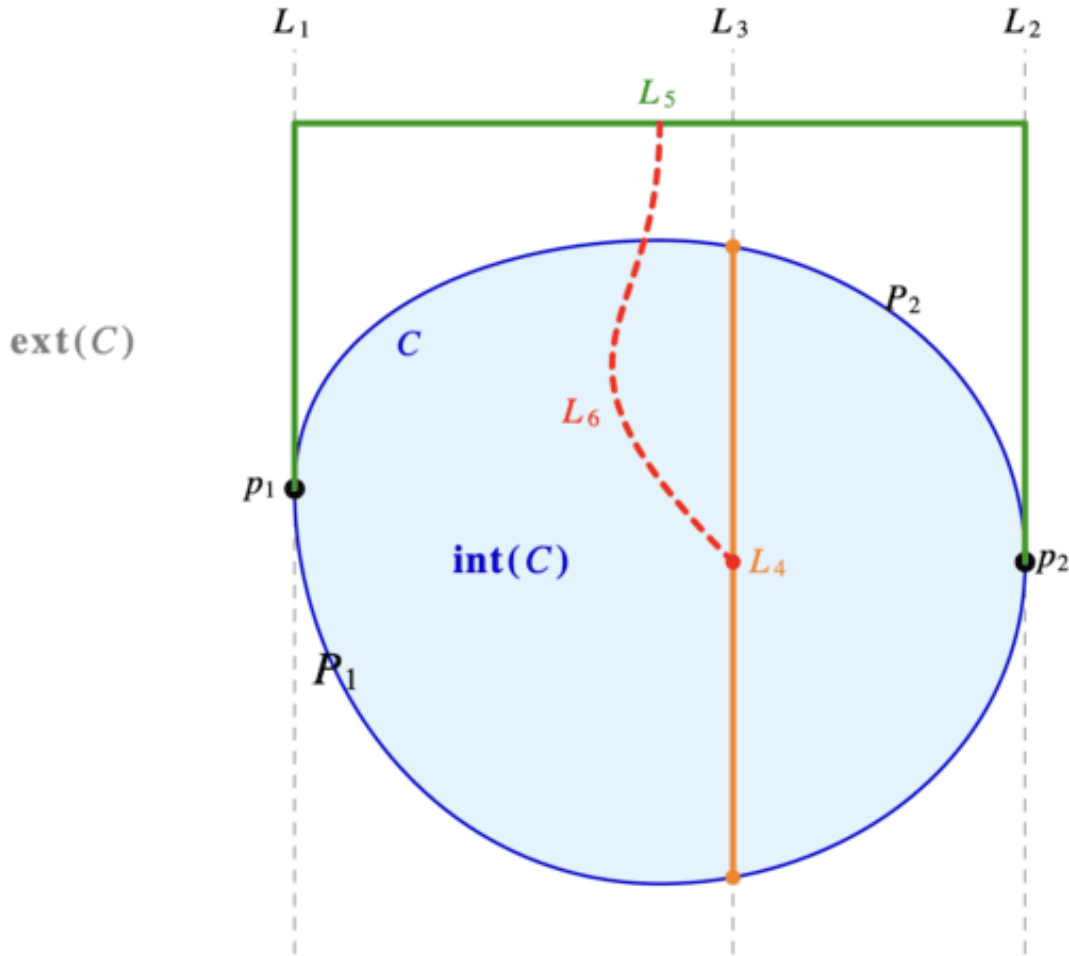
Proof: Clearly $\text{ext}(C)$ is a component of $\mathbb{R}^2 \setminus (C \cup P)$, and analogously $\text{int}(C) \setminus P$ has at most two components. Likewise take a segment in $\text{int}(C)$ crossing P , then the two endpoints are in different

components by the argument in Lemma1. $\square$

Hence for points x, y on the interior of P_1, P_2 , we can't join them without intersecting $C \cup P$.

Theorem3: If C is a simple closed curve in $\mathbb{R}^2$, then $\mathbb{R}^2 \setminus C$ is disconnected.

Proof:



Take a disk covering C , and a point outside this disk. Denote the connected component of this point by $\text{ext}(C)$

Let L_1, L_2 be vertical straight lines intersecting C such that C is entirely in the closed right (resp. left) half plane of L_1 (resp. L_2). Let p_i be the top point on $L_i \cap C$, and P_1, P_2 be the two curves on C connecting p_1, p_2 . Let L_3 be a vertical straight line between L_1, L_2 . Since $P_1 \cap L_3$ and $P_2 \cap L_3$ are compact and disjoint, L_3 contains an interval L_4 joining P_1, P_2 and having only its ends in C . Let L_5 be a polygonal arc from p_1 to p_2 in $\text{ext}(C)$ consisting of segments of L_1, L_2 , and a horizontal straight segment above C . If L_4 is in $\overline{\text{ext}(C)}$, then there is a simple polygonal arc L_6 in $\text{ext}(C)$ from L_4 to L_5 . But then $C \cup L_4 \cup L_5 \cup L_6$ is a planar representation of $K_{3,3}$, a contradiction. Hence the midpoint of L_4 does not lie in $\overline{\text{ext}(C)}$, so $\text{int}(C)$ is nonempty. $\square$

Lemma4: If G is a 2-connected graph and H is a 2-connected subgraph of G , then G can be obtained from H : in each step add a path of G whose endpoints lie in the current graph.
(This is purely combinatorial, a simple induction on $E(G \setminus H)$.)

Corollary5: If Γ is a planar 2-connected graph with $V(\Gamma) \geq 3$ and PL edges, then $\mathbb{R}^2 \setminus \Gamma$ has $|E(\Gamma)| - |V(\Gamma)| + 2$ regions, each having a cycle of Γ as boundary.

Proof: Start from a cycle in Γ , and use Lemma4 and Lemma2.

Proposition6: If P is a simple arc (open) in the plane, then $\mathbb{R}^2 \setminus P$ is connected.

Proof: Let p, q be two points in $\mathbb{R}^2 \setminus P$, and $d > 0$ such that p, q has distance $> 3d$ from P . Since P is the image of a (uniformly) continuous map, we partition P into segments $P_1, \dots, P_k$ such that P_i joins p_i, p_{i+1} , and each point on P_i has distance $< d$ from p_i .

Let d' be the minimum distance between P_i, P_j for $i \leq j - 2$, then $d' \leq d$.

For each $i = 1, \dots, k$, partition P_i into segments $P_{i,1}, \dots, P_{i,k_i}$ such that $P_{i,j}$ joins $p_{i,j}, p_{i,j+1}$ and that each point on $P_{i,j}$ has distance less than $d'/4$ to $p_{i,j}$, and let Γ_i be the graph consisting of the union of squares with side-length $d'/2$ and $p_{i,j}$ as midpoints. Then we can show both p, q are in the outer face of $\Gamma_1 \cup \dots \cup \Gamma_k$, since they are outside $B(p_i, 3d)$ while $\Gamma_i \cup \Gamma_{i+1}$ is inside $B(p_i, 3d)$, and P does not intersect that face. Therefore p, q can be joined by a simple polygonal arc disjoint from P . $\square$

If $C \subset \mathbb{R}^2$ is closed and Ω is a region of $\mathbb{R}^2 \setminus C$, then a point p in C is accessible if for some $q \in \Omega$, there is a PL curve from q to p intersecting C only at p .

If P is any arc of C , then Prop6 shows that there is an accessible point on P . Hence the accessible points on C are dense.

Theorem7: (Jordan Curve Theorem) If C is a simple closed curve in $\mathbb{R}^2$, then $\mathbb{R}^2 \setminus C$ has exactly two connected components, having C as boundary.

Proof: Otherwise suppose q_1, q_2, q_3 are points in distinct components $\Omega_1, \Omega_2, \Omega_3$, and let Q_1, Q_2, Q_3 be disjoint segments of C . Ω_i has a simple PL arc $P_{i,j}$ from q_i to Q_j , and we can assume $P_{i,j} \cap P_{i,j'} = \{q_i\}$ for $j \neq j'$. Clearly $P_{i,j} \cap P_{i',j'} = \emptyset$ when $i \neq i'$, so we obtain a planar representation of $K_{3,3}$ (by adding segments in Q_i), a contradiction. Hence $\mathbb{R}^2 \setminus C$ has precisely two components $\text{ext}(C)$ and $\text{int}(C)$. Since the accessible points are dense, C is the boundary of $\text{ext}(C)$ and $\text{int}(C)$. $\square$

Proof by General Topology

Theorem1: If C is a PL (Piecewise Linear) closed curve in $\mathbb{R}^2$, then $\mathbb{R}^2 \setminus C$ has exactly two connected components.

Proof: By taking $a \in \mathbb{R}^2 \setminus C$, and for each $x \in \mathbb{R}^2 \setminus C$, consider the parity of number of intersections of C with $[a, x]$ (segment from a to x), if $[a, x]$ passes a vertex or covers an edge, we view it as two intersections. We know that $\mathbb{R}^2 \setminus C$ is disconnected, since the two sets of points with even and odd parity are open and disjoint. If q_1, q_2, q_3 lie in distinct components, take a small disk D such that $D \cap C$ is a line segment. Then take a tubular neighborhood of C , and walking along this neighborhood we obtain polygonal arcs in $\mathbb{R}^2 \setminus C$ sending q_i into D , a contradiction. $\square$

Lemma2: Suppose $E \subset \mathbb{S}^1$ such that $\forall x, y \in E, |x - y| < \sqrt{3}$, then there is a closed arc of $\mathbb{S}^1$ with length $\leq \frac{2\pi}{3}$ that covers E .

Proof: Assume E has at least 3 points. Take $a, b \in E$, and WLOG assume $a = e^{2\pi i/3}, b = e^{2\pi i\theta}$ with $\theta \in (1/3, 2/3)$. Then for any $c = e^{2\pi i\phi} \in E$, since $|a - c|, |b - c| < \sqrt{3}$, we can take $\phi \in (\theta - 1/3, 2/3) \subset (0, 2/3)$. Let $\psi_1 = \inf\{\phi \in (\theta - 1/3, 2/3) : e^{2\pi i\phi} \in E\}$ and $\psi_2 = \sup\{\phi \in (\theta - 1/3, 2/3) : e^{2\pi i\phi} \in E\}$, we show that $\psi_2 - \psi_1 \leq 1/3$. Otherwise, take ϕ_1, ϕ_2 such that $e^{2\pi i\phi_j} \in E$ and $\phi_2 - \phi_1 > 1/3$. Since $\phi_j \in (0, 2/3)$, $\phi_1 + 1 - \phi_2 > 1 - 2/3 = 1/3$, contradicting $|e^{2\pi i\phi_1} - e^{2\pi i\phi_2}| < \sqrt{3}$. Hence we obtain a arc with length $\leq \frac{2\pi}{3}$ covering E . $\square$

Theorem3: Any Jordan curve can be approximated by PL arcs: given $f : \mathbb{S}^1 \rightarrow \mathbb{R}^2$ and its image C , for any $\varepsilon > 0$ there exists $f' : \mathbb{S}^1 \rightarrow \mathbb{R}^2$ with image C' such that C' is PL and $\|f - f'\|_\infty < \varepsilon$.

Proof: f is uniformly continuous on the compact set $\mathbb{S}^1$, so take $\varepsilon_1 > 0$ such that $|x - y| < \varepsilon_1 \implies |f(x) - f(y)| < \varepsilon/2$. Take $\varepsilon_2 > 0$ such that $|f(x) - f(y)| < \varepsilon_2 \implies |x - y| < \min\{\varepsilon_1, \sqrt{3}\}$ (since $f : \mathbb{S}^1 \rightarrow C$ is a homeomorphism).

Let $\delta = \min\{\varepsilon/2, \varepsilon_2\}$, we cover C using finitely many cubes $Q_1, \dots, Q_N$ of diameter δ , such that their interiors are disjoint. By Lemma2, the minimal arc A_1 that covers $f^{-1}(Q_1)$ has length $\leq \frac{2\pi}{3}$, so $A_1 \neq \mathbb{S}^1$. Let f_1 be f with the $[a, b]$ arc turned into a line, where $A_1 = \{e^{2\pi it} : a \leq t \leq b\}$, and let $C_1 = f_1(\mathbb{S}^1)$. This process pulls $f(A_1)$ into a straight line. Note that $f_1^{-1}(Q_i) \subset f^{-1}(Q_i)$. Repeat this process, by taking A_2 covering $f_1^{-1}(Q_2)$, and straightening $f_1(A_2)$ to obtain f_2 . We obtain $f_N : \mathbb{S}^1 \rightarrow \mathbb{R}^2$ and a Jordan polygon $C_n = f_n(\mathbb{S}^1)$. We show that $|f(x) - f_n(x)| < \varepsilon$:

Suppose $x \in \mathbb{S}^1$, and $f(x) \neq f_n(x)$, take j such that $f_n(x) = f_j(x) \neq f_{j-1}(x)$, then $x \in A_j$. Suppose the endpoints of the arc A_j are y, z , then by definition $f_j(y) = f(y), f_j(z) = f(z)$, so

$$|f(x) - f_n(x)| \leq |f(x) - f(y)| + |f_j(x) - f_j(y)| \leq |f(x) - f(y)| + \delta.$$

Since $|f(z) - f(y)| \leq \delta \leq \varepsilon_2$, we have $|z - y| \leq \varepsilon_1$. By definition the length of A_j is less than $\frac{2\pi}{3}$, so $|x - y| < |z - y| < \varepsilon_1$. Hence $|f(x) - f(y)| \leq \varepsilon/2$ and $|f(x) - f_n(x)| < \varepsilon$. $\square$

Theorem4: Suppose C is a Jordan polygon given by $f : \mathbb{S}^1 \rightarrow \mathbb{R}^2$, then there is an open disk in $\text{int}(C)$, such that its boundary meets C at points $f(a), f(b)$ with $|a - b| \geq \sqrt{3}$.

Theorem5: If C is a Jordan curve, then $\mathbb{R}^2 \setminus C$ is disconnected.

Proof: Clearly $\mathbb{R}^2 \setminus C$ has an unbounded component, we show that $\mathbb{R}^2 \setminus C$ has a bounded component. Take a sequence of PL arcs C_n that approximate C , such that C_n is given by $f_n : \mathbb{S}^1 \rightarrow \mathbb{R}^2$ and $|f(x) - f_n(x)| \leq \varepsilon_n$, $\lim_{n \rightarrow \infty} \varepsilon_n = 0$.

By Lemma4, in the interior of each C_n , there is a circle S_n , containing points $f_n(a_n)$ and $f_n(b_n)$ from C_n , such that $|a_n - b_n| \geq \sqrt{3}$. Suppose the center of S_n is z_n , then take a converging subsequence of z_n . WLOG assume z_n converges, and the limit is z .

We show that: for large n , z and z_n belong to the same component of $\mathbb{R}^2 \setminus C_n$, namely, the bounded one $\text{int}(C_n)$. Since f is a homeomorphism, there exists $\delta > 0$ such that $|f(x) - f(y)| < \delta \implies |x - y| < \sqrt{3}$. Take N such that $n \geq N \implies \varepsilon_n < \delta/4$, then for $n \geq N$, $|f_n(x) - f_n(y)| > \delta/2$ since $|x - y| \geq \sqrt{3}$. Hence $\text{diam} S_n > \delta/2$ so $d(z_n, C_n) > \delta/4$. For large n , $|z - z_n| < \delta/4$, so z and z_n belong to the same component of $\mathbb{R}^2 \setminus C_n$, which is clearly the bounded component of $\mathbb{R}^2 \setminus C_n$.

Then we show that z cannot belong to the unbounded component of $\mathbb{R}^2 \setminus C$, hence $\mathbb{R}^2 \setminus C$ is disconnected: suppose the contrary, then take a continuous path $g : [0, 1] \rightarrow \mathbb{R}^2 \setminus C$ taking z to $\text{ext}(C_n)$. Since $[0, 1]$ is compact, $d(\text{img}, C) = \delta > 0$. For large n , $|f(x) - f_n(x)| < \delta/2$, so $d(\text{img}, C_n) > \delta/2$ which implies $z \in \text{ext}(C_n)$ for large n , a contradiction.

Lemma6: Suppose C is a Jordan polygon, a, b belong to the same component X of $\mathbb{R}^2 \setminus C$, and exists $\delta > 0$ such that $d(\{a, b\}, C) > \delta$. Further suppose for any segment $[c, d] \subset X \cup C$ of length less than 2δ , a, b belong to the same connected component of $X \setminus [c, d]$. Prove that there is a path $g : [0, 1] \rightarrow X$ in X connecting a, b such that $d(\text{img}, C) \geq \delta$.

Proof: WLOG assume that $a, b \in \text{int}(C)$. The dragging of the disk is divided into three stages:

(I) There exists a point $\alpha \in C$ such that $d(a, \alpha) = d(a, C)$. Let α_1 be a point on the line segment $[a, \alpha]$ such

that the length of the segment $[\alpha_1, \alpha]$ is δ . Since $d(a, \alpha) = d(a, C) > \delta$, the center of an open disk of radius δ can be dragged from a to α_1 along the line segment $[a, \alpha_1]$ without the disk ever touching C .

(II) At the start of stage (II), we have $d(\alpha_1, C) = \delta$. We now drag this open disk "forward" while maintaining its tangency to C at point α . There are two cases to consider when the point of tangency passes a vertex of C :

- (i) If the interior angle subtended at the vertex is greater than π , the disk "rolls" around the vertex until it becomes tangent to the next edge. The trajectory of the disk's center near the vertex is a circular arc of radius δ , which lies within the interior region. This arc is tangent to the two straight paths traced by the center before and after passing the vertex.
- (ii) If the interior angle subtended at the vertex is less than π , the disk will reach a position tangent to the next edge before its boundary hits the vertex. In this case, the center simply continues its movement along the next tangent line.

The term "dragging forward" is interpreted as follows: starting from position α_1 , there are two possible directions to drag the disk tangent to a straight segment of C . Dragging forward means following one of these directions such that, after a small movement, the extension of the line segment from the center to the point of tangency forms a chord λ that divides $C \setminus \lambda$ into two components, separating a and b into different components. According to the assumption of the lemma, no such chord can hinder the disk. Thus, we can unimpededly drag the disk to a position where the perpendicular extension from its center to the point of tangency with C coincides with the endpoints of the target chord.

(III) This stage is analogous to stage (I), but performed in reverse, moving from the final tangent position to point b . Based on the same reasoning used in stage (I), this part of the path will also be free of obstacles.

Theorem7: If C is a Jordan curve, then $\mathbb{R}^2 \setminus C$ has at most 2 components.

Proof: Otherwise if $\mathbb{R}^2 \setminus C$ has three or more components, suppose p, q, r are points belonging to different components, and let $\varepsilon = d(\{p, q, r\}, C) > 0$. Take a sequence of Jordan polygons $\{f_n : S^1 \rightarrow C_n\}$ such that $\lim_{n \rightarrow \infty} \|f - f_n\| = 0$. Then for large n , $\|f - f_n\| < \varepsilon/2$ so $d(\{p, q, r\}, C_n) > \varepsilon/2$. Since $\mathbb{R}^2 \setminus C_n$ has two components, we can assume p, q belong to the same component X_n of C_n (otherwise take a sub-sequence that has this property).

If for $\delta \in (0, \varepsilon)$ and infinitely many n , there is a path g_n in X_n connecting p, q such that $d(\text{Img}_n, C_n) \geq \delta$, then for large n , $\|f - f_n\| < \delta/2$ so $d(\text{Img}_n, C) > \delta/2$. Hence p, q belong to the same component of $\mathbb{R}^2 \setminus C$, a contradiction, so no such δ exists.

Consequently, the paths connecting p and q in X_n must approach C as n increases. Let $\delta_n = d(g_n([0, 1]), C_n)$. By selecting a suitable subsequence, we can assume the conditions of the proof are satisfied. According to Proposition 7.10.5, for each n , there exists a set $\Gamma_n \subset X_n$ with small diameter such that p and q are separated by Γ_n . Let $f_n(a_n)$ and $f_n(b_n)$ be the endpoints of Γ_n . Since $\delta_n \rightarrow 0$, it follows that $\lim_{n \rightarrow \infty} |f_n(b_n) - f_n(a_n)| = 0$. Given that $f_n \rightarrow f$ uniformly, we have $\lim_{n \rightarrow \infty} (f(b_n) - f(a_n)) = 0$. Since f is a homeomorphism (and thus uniformly continuous), it follows that $\lim_{n \rightarrow \infty} (b_n - a_n) = 0$. Because p and q belong to two different connected components of $X_n \setminus \Gamma_n$, for infinitely many n , p and q are separated by one of the arcs of C_n between $f_n(a_n)$ and $f_n(b_n)$. Let A_n denote the shorter arc on S^1 with endpoints a_n and b_n . Since $f_n \rightarrow f$ uniformly and f is continuous, the diameter of $f_n(A_n)$ tends to zero. Furthermore, since the diameter of Γ_n is not large, the set $K_n = f_n(A_n) \cup \Gamma_n$ satisfies $\text{diam}(K_n) \rightarrow 0$. In summary, there exists a strictly increasing sequence of positive integers $\{n_i\}$ ($i = 1, 2, \dots$) such that $p \in K_{n_i}$, $i = 1, 2, \dots$.

When i is sufficiently large, $\text{diam}(K_{n_i}) < e$ (where e is the positive constant $d(\{p, q, r\}, C) = e$ defined at the beginning of the proof). This implies $d(p, C) < e$, which contradicts the fact that $d(p, C) \geq e$. $\square$